

PREECLAMPSIA DIFFERENTIAL GENE EXPRESSION ANALYSIS

A Computational Transcriptomic Analysis of Normal and Preeclamptic Placental Samples

Dataset: GSE190971

Analysis Platform: GEO2R and Python

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ABSTRACT

Preeclampsia is a complex hypertensive disorder of pregnancy that can significantly affect maternal and fetal health. The primary objective of this project was to investigate differential gene expression patterns between normal and preeclamptic placental samples using a computational approach. The publicly available transcriptomic dataset GSE190971 was analyzed, with initial differential expression results obtained through GEO2R. The resulting data were subsequently imported, cleaned, filtered, summarized, and visualized in a Python environment.

For statistical filtering, an adjusted p-value (padj) threshold of < 0.05 and an absolute log2 fold-change ($|\log_2FC|$) threshold of ≥ 1 were applied. A total of 18,765 genes were analyzed, of which 854 met the predefined criteria for significant differential expression. Based on the comparison convention used in the project, 208 genes were classified as showing higher expression in the Normal group, whereas 646 genes were classified as showing higher expression in the Preeclampsia group.

A volcano plot was generated to visualize both the magnitude of differential expression and statistical significance. Selected genes, including LEP, DIO2, EGLN3-AS1, HTRA4, and EGLN3, were highlighted as candidates for further investigation. Overall, this analysis provides a systematic exploration of placental transcriptomic differences associated with preeclampsia and identifies candidate genes for future functional enrichment, pathway analysis, independent dataset validation, and experimental investigation.

Keywords: Preeclampsia, Placenta, Differential Gene Expression, Transcriptomics, GEO2R, Python, Volcano Plot

1. INTRODUCTION

1.1 Background

Preeclampsia is a pregnancy-associated multisystem disorder characterized by new-onset hypertension, often accompanied by maternal organ dysfunction and other clinical complications. Its pathophysiology is complex and multifactorial, involving disturbances in placental development, vascular regulation, angiogenic balance, hypoxia-related responses, inflammatory signaling, and maternal systemic adaptation. The placenta plays a central role in the development of the disorder, as abnormal placentation and altered placental signaling are considered important contributors to disease progression.

Gene expression analysis provides a valuable approach for investigating molecular differences between healthy and disease-associated conditions. RNA sequencing (RNA-seq)-based transcriptomic datasets enable the simultaneous analysis of thousands of genes. Differential expression analysis can identify statistically meaningful changes that may reveal altered biological processes, molecular pathways, and candidate genes associated with disease pathophysiology.

1.2 Rationale of the Study

Preeclampsia is a clinically heterogeneous disorder, making it difficult to explain its complex pathophysiology through a single gene or molecular pathway. Public transcriptomic data re-analysis provides an accessible and reproducible strategy for exploring disease-associated molecular patterns. In this project, GSE190971 was selected to investigate placental transcriptomic differences between normal and preeclamptic pregnancies.

The purpose of this project was not to claim independent wet-lab validation or establish causality. Rather, the study aimed to identify, filter, classify, and visualize candidate differentially expressed genes (DEGs) that may warrant further biological and functional investigation.

2. AIM AND OBJECTIVES

2.1 Aim

The aim of this project was to identify, filter, summarize, and visualize differential gene expression patterns between normal and preeclamptic placental samples using GEO2R-derived results and a Python-based computational workflow.

2.2 Objectives

- Obtain differential expression results for GSE190971 using GEO2R.

- Import and clean gene-level results in Python.
- Identify significant DEGs using $\text{padj} < 0.05$ and $|\log_2\text{FC}| \geq 1$.
- Classify genes according to project-defined contrast direction.
- Calculate summary statistics.
- Generate a labeled volcano plot.
- Identify candidates for further interpretation and validation.

3. DATASET DESCRIPTION

In this project, the publicly available dataset GSE190971 from the NCBI Gene Expression Omnibus (GEO) was used. The dataset is related to placental transcriptomics in the context of normal and preeclamptic pregnancies. Within the GEO2R interface, selected samples were assigned to defined biological groups for differential expression analysis. The downloaded results table contained gene-level statistical and annotation fields, including GeneID, adjusted p-value (padj), p-value, log2FoldChange, baseMean, Symbol, and Description.

Important methodological note: The final interpretation of differential expression results depends on the direction of the comparison. In this report, gene-direction labels are described according to the Normal versus Preeclampsia contrast convention used in the project notebook. To ensure reproducibility and accurate biological interpretation, the exact sample group assignments and contrast direction should be documented and preserved alongside the analysis notebook.

4. MATERIALS AND METHODS

4.1 Software and Resources

- NCBI GEO
- GEO2R
- Google Colab
- Python
- pandas
- NumPy
- Matplotlib
- Microsoft Excel / CSV files

4.2 Analysis Workflow

The GSE190971 accession was first accessed through GEO2R, where selected normal and preeclampsia samples were assigned to their respective groups. The differential expression output was downloaded in spreadsheet format and imported into Google Colab for Python-based processing. Following data import, the dataset structure and relevant statistical columns were inspected, and cleaning was performed to address missing values in key variables.

To identify significantly differentially expressed genes, two criteria were applied simultaneously: adjusted p-value ($\text{padj} < 0.05$) and absolute log2 fold change ($|\log_2\text{FoldChange}| \geq 1$). The adjusted p-value threshold was used to account for multiple testing, whereas the fold-change threshold retained genes with substantial expression differences. Filtered genes were divided into direction-specific subsets according to the sign of log2 fold change and the predefined contrast convention.

4.3 Statistical Filtering Criteria

Parameter	Threshold	Purpose
Adjusted p-value	$\text{padj} < 0.05$	Statistical significance after adjustment
Fold change	$ \log_2\text{FC} \geq 1$	Substantial expression difference
Positive direction	$\log_2\text{FC} \geq 1$	Higher in project-defined Normal direction

Parameter	Threshold	Purpose
Negative direction	$\log_2\text{FC} \leq -1$	Higher in project-defined Preeclampsia direction

4.4 Data Visualization

A volcano plot was generated with log2 fold change on the x-axis and -log10(adjusted p-value) on the y-axis. This visualization enables simultaneous assessment of expression-change magnitude and statistical significance. Threshold lines were added, direction-specific significant genes were displayed as separate categories, and selected genes were highlighted using gene-symbol labels.

5. RESULTS

5.1 Summary of Differential Expression

Following application of the predefined filtering criteria, 854 significant DEGs were identified. Based on the contrast convention used in this project, 208 genes exhibited higher expression in the Normal group, whereas 646 genes showed higher expression in the Preeclampsia group. The significant DEG set displayed negative mean and median log2 fold-change values, indicating an overall asymmetry in expression direction under the selected contrast convention. This observation does not demonstrate direct biological causation; it represents a dataset-specific differential expression pattern.

5.2 Volcano Plot Interpretation

The generated volcano plot provides a simultaneous visual representation of differential expression magnitude and statistical significance. Most genes cluster around the central region, where log2 fold-change values are relatively small. Genes beyond the predefined fold-change thresholds exhibit larger expression differences between the compared groups.

The y-axis represents -log10(adjusted p-value), such that genes positioned higher on the plot have smaller adjusted p-values and stronger statistical evidence of differential expression. Genes meeting both criteria were displayed as direction-specific significant categories according to the project-defined contrast convention. Among highlighted candidates, LEP exhibited a particularly strong differential signal. DIO2, EGLN3-AS1, HTRA4, EGLN3, and other selected candidates also showed notable patterns warranting further investigation.

Figure 1. Labeled volcano plot showing differential gene expression between Normal and Preeclampsia groups. The x-axis represents log2 fold change, while the y-axis represents -log10(adjusted p-value). Dashed threshold lines indicate predefined statistical significance and fold-change criteria.

5.3 Selected Candidate Genes

Gene	Observed Role	Interpretive Note
LEP	Strong differential signal; highly significant	Follow-up in placental function and metabolic/endocrine regulation.
DIO2	Strongly altered candidate	Investigate thyroid hormone metabolism and placental physiology.
EGLN3-AS1	Differentially expressed candidate	Further annotation is required to clarify regulatory relevance.
HTRA4	Strongly altered candidate	Investigate in placental biology and preeclampsia literature.
EGLN3	Differentially expressed candidate	Further analysis may clarify oxygen-sensing or hypoxia-related involvement.

6. DISCUSSION

The present analysis indicates substantial transcriptomic differences between normal and preeclamptic placental samples. Identification of 854 filtered DEGs suggests that multiple molecular processes may be altered in disease-associated placenta. However, DEG count should not be interpreted as direct evidence of disease mechanism, because expression differences can be influenced by sample composition, clinical heterogeneity, gestational variables, technical factors, and other confounders.

The direction-specific imbalance, in which the number of genes classified as higher in the Preeclampsia group exceeded the number classified as higher in the Normal group, represents a notable dataset-level pattern. To understand this pattern biologically, Gene Ontology enrichment, KEGG or Reactome pathway analysis, gene-set enrichment, and network analysis would be valuable next steps.

Highly significant candidates such as LEP warrant further attention; however, statistical significance alone does not establish biomarker validity. Potential biomarker claims require independent cohorts, consistency of effect size, diagnostic performance assessment, adjustment for clinical confounders, and preferably experimental validation. DIO2, HTRA4, EGLN3, and regulatory transcripts should also be evaluated using existing literature and external transcriptomic datasets.

7. LIMITATIONS

- Secondary public data were analyzed; no independent wet-lab validation was performed.
- Output depends on GEO2R sample grouping and comparison settings.
- Direction interpretation is linked to group coding and must be documented accurately.
- Potential clinical confounders were not independently modeled in the Python filtering stage.
- Differential expression shows association, not causation.
- Candidate genes require independent and experimental validation.

8. FUTURE DIRECTIONS

- Gene Ontology enrichment analysis.
- KEGG or Reactome pathway enrichment.
- Protein-protein interaction network analysis.
- Validation in independent preeclampsia datasets.
- Heatmaps and sample-level expression plots when suitable data are available.
- Multivariable analysis with clinical metadata.
- ROC analysis only with appropriate sample-level data and study design.
- qPCR or other experimental validation of key findings.

9. CONCLUSION

In this project, GEO2R-derived differential expression results from the GSE190971 placental transcriptomic dataset were analyzed using a Python-based computational workflow. After applying $\text{padj} < 0.05$ and $|\log_2\text{FC}| \geq 1$, 854 significant DEGs were identified. Based on the contrast convention used in the project, 208 genes exhibited higher expression in the Normal group and 646 genes showed higher expression in the Preeclampsia group.

The labeled volcano plot summarized expression magnitude and statistical significance and highlighted selected candidates, including LEP, DIO2, EGLN3-AS1, HTRA4, and EGLN3. These findings should be interpreted as dataset-specific associations rather than evidence of direct causation or validated biomarker performance.

Overall, the project demonstrates how public transcriptomic data, statistical filtering, and Python-based visualization can be integrated into a reproducible workflow to investigate disease-associated molecular patterns. Functional enrichment, pathway-level interpretation, independent dataset validation, clinical metadata, and experimental confirmation would strengthen future biological interpretation.

10. REPRODUCIBILITY AND REPOSITORY CONTENTS

- Preeclampsia_Differential_Gene_Expression_Analysis.ipynb — Python notebook
- GSE190971_GEO2R_results.xlsx — GEO2R-derived results
- filtered_DEGs_padj0.05_log2FC1.csv — Filtered DEGs
- genes_higher_in_normal.csv — Normal-higher subset
- genes_higher_in_preeclampsia.csv — Preeclampsia-higher subset
- summary_statistics.csv — Summary metrics

- volcano_plot_labeled.png — Labeled volcano plot
- README.md — Repository documentation

APPENDIX A: SHORT WORKFLOW SUMMARY

GSE190971 selection → Normal and Preeclampsia sample grouping in GEO2R → Differential expression result download → XLSX import into Google Colab → Data inspection and cleaning → $\text{padj} < 0.05$ filtering → $|\log_2\text{FC}| \geq 1$ filtering → Direction-based gene separation → Summary statistics → Labeled volcano plot → CSV export → GitHub documentation.

REFERENCED SCIENTIFIC BACKGROUND

Preeclampsia is a complex multisystem pregnancy disorder characterized by hypertension and may be accompanied by maternal organ dysfunction. Current evidence indicates that abnormal placentation, placental stress, angiogenic imbalance, and maternal endothelial dysfunction are important components of its pathophysiology (Phipps et al., 2019; Rana et al., 2019, 2022).

The placenta plays a central role in the pathogenesis of preeclampsia. Impaired trophoblast invasion and abnormalities in spiral artery remodeling may disrupt placental perfusion and contribute to placental dysfunction. Subsequently, placenta-derived circulating factors may promote systemic maternal vascular and endothelial abnormalities (Phipps et al., 2019; Rana et al., 2019; Staff, 2019).

Transcriptomic profiling provides a systematic approach for investigating differences in gene expression between normal and disease-associated placental tissues. Previous placental gene expression studies have demonstrated alterations across multiple molecular pathways and gene expression programs (Brew et al., 2016; Leavey et al., 2016).

In this project, GSE190971 was analyzed. According to the GEO accession record, RNA sequencing was performed using an Illumina HiSeq platform, while the reported processing workflow included HISAT2 for alignment and featureCounts for gene-level quantification (NCBI Gene Expression Omnibus, 2023).

The Gene Expression Omnibus is a public repository for functional genomics data, whereas GEO2R provides a web-based interface for comparing gene expression profiles between user-defined sample groups (Edgar et al., 2002; NCBI Gene Expression Omnibus, n.d.). DESeq2 is a widely used statistical framework for count-based RNA-seq differential expression analysis (Love et al., 2014).

Published evidence has reported placenta-associated expression of HTRA4 and altered HTRA4 levels in early-onset preeclampsia (Singh et al., 2015). Candidate genes identified here should therefore be regarded as hypothesis-generating findings rather than direct causal factors or clinically validated biomarkers.

CITATION-GUIDED INTERPRETATION OF PROJECT RESULTS

In the present analysis, 18,765 genes were evaluated, of which 854 met the predefined criteria for significant differential expression. Based on the comparison direction used in the project notebook, 208 genes exhibited higher expression in the Normal group, whereas 646 genes showed higher expression in the Preeclampsia group.

Adjustment of p-values is important in high-dimensional transcriptomic analyses because thousands of statistical tests are performed simultaneously. The Benjamini-Hochberg procedure represents a foundational approach for controlling the false discovery rate in multiple-hypothesis testing (Benjamini & Hochberg, 1995).

Strongly shifted candidates, including LEP, DIO2, EGLN3-AS1, and HTRA4, may warrant further investigation. However, computational differential expression alone cannot establish causation, diagnostic utility, or clinical biomarker validity. Independent cohorts, appropriate clinical design, confounder adjustment, and complementary experimental validation are required (Brew et al., 2016; Leavey et al., 2016; Rana et al., 2019).

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